INTERSECTION THEORY IN ALGEBRAIC GEOMETRY: FROM FIRST PRINCIPLES TO FULTON–MACPHERSON

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Abstract. These notes develop intersection theory in algebraic geometry with complete proofs of all major results. Starting from scratch with the Nullstellensatz and the Noetherian decomposition theorem, we prove the algebra–geometry anti-equivalence, the regularity criterion for smoothness, projectivity of global functions, the Weil–Cartier divisor theory, the definition and basic properties of Chow groups, Kleiman's proof of the moving lemma via the action of a connected algebraic group, Serre's intersection multiplicity formula via Tor, the construction of the deformation space and the Verdier specialization map, Fulton's refined Gysin homomorphism, the excess intersection formula, Grothendieck–Riemann–Roch (for embeddings and projections), Chow's theorem on algebraicity of analytic sets in projective space, and Serre's GAGA theorem. Bloch's formula connecting Chow groups to K-theory cohomology is derived via the Gersten resolution. Every theorem is accompanied by a complete proof or a proof with all essential steps made explicit.

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1. The Nullstellensatz and the algebra–geometry dictionary

1.1. **Setup and statement.** Throughout this section k is an algebraically closed field. We write $k[x] = k[x_1, \dots, x_n]$ for the polynomial ring. For an ideal $I \subseteq k[x]$, define the *vanishing set* $V(I) = \{a \in k^n : f(a) = 0 \ \forall f \in I\}$. For a subset $X \subseteq k^n$, define the *vanishing ideal* $I(X) = \{f \in k[x] : f|_X \equiv 0\}$.

Theorem 1.1 (Hilbert Nullstellensatz)**.** *For any ideal $I \subseteq k[x_1, \dots, x_n]$ with k algebraically closed,*

$$I(V(I)) = \sqrt{I} := \{f \in k[x] : f^m \in I \text{ for some } m \geq 1\}.$$

We prove this in two steps.

Theorem 1.2 (Weak Nullstellensatz)**.** *If $I \subsetneq k[x_1, \dots, x_n]$ is a proper ideal and k is algebraically closed, then $V(I) \neq \varnothing$.*

Proof. Let $\mathfrak{m} \supseteq I$ be a maximal ideal (which exists by Zorn's lemma, since $I \subsetneq k[x]$). Consider the quotient field $K = k[x]/\mathfrak{m}$. Since $\mathfrak{m}$ is maximal, K is a field. The inclusion $k \hookrightarrow k[x] \twoheadrightarrow K$ makes K a field extension of k. Each generator $\bar{x}_i \in K$ satisfies a polynomial relation over k (since K is the quotient of a finitely generated k-algebra), so K/k is an algebraic extension. Since k is algebraically closed, every algebraic extension is trivial, hence $K = k$.

Therefore the composition $k[x] \twoheadrightarrow k[x]/\mathfrak{m} \xrightarrow{\sim} k$ sends each x_i to some element $a_i \in k$. The point $a = (a_1, \dots, a_n)$ satisfies $f(a) = 0$ for all $f \in \mathfrak{m} \supseteq I$, so $a \in V(I)$. □

Proof of Theorem 1.1 (Rabinowitsch trick). The inclusion $\sqrt{I} \subseteq I(V(I))$ is clear: if $f^m \in I$ and $a \in V(I)$, then $f(a)^m = 0$ in k, so $f(a) = 0$.

For the other inclusion, let $g \in I(V(I))$; we must show $g \in \sqrt{I}$. Consider the ideal

$$J = I \cdot k[x_1, \dots, x_n, t] + (1 - tg) \subseteq k[x_1, \dots, x_n, t].$$

We claim $J = (1)$. If not, then by the Weak Nullstellensatz applied to J in $n+1$ variables, there would exist $(a, b) \in k^{n+1}$ with $f(a) = 0$ for all $f \in I$ and $1 - b\,g(a) = 0$. But $a \in V(I)$ forces $g(a) = 0$ (since $g \in I(V(I))$), giving $1 = 0$, a contradiction. So $J = (1)$.

Write $1 = \sum_j h_j f_j + q(1 - tg)$ where $f_j \in I$ and $h_j, q \in k[x, t]$. Substituting $t = 1/g$ (in the fraction field $k(x)[t]/(tg - 1)$) gives

$$1 = \sum_j h_j(x, 1/g) f_j(x).$$

Multiplying through by a sufficiently high power g^N clears denominators and yields $g^N \in I$, so $g \in \sqrt{I}$. □

Corollary 1.3. *The operations $I \mapsto V(I)$ and $X \mapsto I(X)$ define mutually inverse order-reversing bijections*

$$\{\textit{radical ideals in } k[x]\} \leftrightarrow \{\textit{algebraic sets in } k^n\}.$$

Under this bijection: prime ideals $\leftrightarrow$ irreducible algebraic sets; maximal ideals $\leftrightarrow$ points.

Proof. That $I(V(I)) = \sqrt{I}$ for all ideals I is exactly Theorem 1.1. For a radical ideal $I = \sqrt{I}$, this gives $I(V(I)) = I$. Conversely, $V(I(X)) = X$ for any algebraic set (since $X = V(S)$ for some S and $I(V(S)) = \sqrt{\langle S \rangle} \supseteq S$ implies $V(I(X)) \subseteq V(S) = X$, with the reverse inclusion trivial). This establishes the bijection.

For the second statement: $V(I)$ is irreducible iff $I(V(I)) = \sqrt{I}$ is prime. Indeed, X is irreducible iff $X \neq Z_1 \cup Z_2$ for closed $Z_i \subsetneq X$, iff $I(X) \neq I(Z_1) \cap I(Z_2)$ for $I(Z_i) \supsetneq I(X)$, iff $I(X)$ has no non-trivial factorization as intersection of larger ideals, which for radical ideals is equivalent to being prime. Points correspond to maximal ideals since $V(\mathfrak{m}_a) = \{a\}$ for $\mathfrak{m}_a = (x_1 - a_1, \ldots, x_n - a_n)$. □

1.2. Noetherian decomposition into irreducible components.

Theorem 1.4. *Every algebraic set $X \subseteq k^n$ decomposes uniquely as*

$$X = X_1 \cup \cdots \cup X_r$$

where the X_i are irreducible algebraic sets and $X_i \not\subseteq X_j$ for $i \neq j$ (the irreducible components*).*

Proof. **Existence.** Define a topological space to be *Noetherian* if every descending chain of closed subsets stabilizes. The Zariski topology on k^n is Noetherian because a descending chain $V(I_1) \supseteq V(I_2) \supseteq \cdots$ gives an ascending chain of radical ideals $\sqrt{I_1} \subseteq \sqrt{I_2} \subseteq \cdots$ in the Noetherian ring $k[x]$ (Noetherian by Hilbert's basis theorem), which stabilizes.

In any Noetherian topological space, every closed set is a finite union of irreducibles. Suppose not; among all closed sets without such a decomposition, by Noetherian induction take a minimal one X. Then X is not irreducible, so $X = Z_1 \cup Z_2$ with $Z_i \subsetneq X$ closed. By minimality of X, each Z_i is a finite union of irreducibles, so $X = Z_1 \cup Z_2$ is too – contradiction.

Uniqueness. Given $X = \bigcup_i X_i = \bigcup_j X'_j$, for each i we have $X_i = \bigcup_j (X_i \cap X'_j)$. Since X_i is irreducible, $X_i = X_i \cap X'_j$ for some j, i.e., $X_i \subseteq X'_j$. By symmetry $X'_j \subseteq X_k$ for some k, giving $X_i \subseteq X_k$, so $i = k$ and $X_i = X'_j$. Hence the decompositions coincide after reordering. □

1.3. The coordinate ring and the anti-equivalence.

Definition 1.5. An *affine variety* is an irreducible algebraic set $X \subseteq k^n$. Its *coordinate ring* is $k[X] = k[x_1, \ldots, x_n]/I(X)$, a finitely generated integral domain.

Theorem 1.6 (Anti-equivalence)**.** *The functor $\Gamma : X \mapsto k[X]$, $(\varphi : X \to Y) \mapsto (\varphi^* : k[Y] \to k[X])$ defined by $\varphi^*(f) = f \circ \varphi$ is a contravariant equivalence:*

$$\{\textit{affine varieties}\}^{\mathrm{op}} \underset{\mathrm{Specm}}{\overset{\Gamma}{\rightleftarrows}} \{\textit{f.g. integral } k\textit{-algebras}\}$$

Proof. **Fully faithful.** For affine varieties $X \subseteq k^n$ and $Y \subseteq k^m$, we must show

$$\{\text{morphisms } X \to Y\} \xrightarrow{\sim} \mathrm{Hom}_{k\text{-alg}}(k[Y], k[X]).$$

A morphism $\varphi = (\varphi_1, \ldots, \varphi_m) : X \to Y$ with $\varphi_i \in k[X]$ determines $\varphi^* : k[Y] \to k[X]$ by $\overline{y}_j \mapsto \varphi_j$, where $\overline{y}_j$ is the image of y_j in $k[Y]$. This is well-defined since $\varphi(X) \subseteq Y$: for any $f \in I(Y)$, $f(\varphi_1, \ldots, \varphi_m)(x) = f(\varphi(x)) = 0$ because $\varphi(x) \in Y$.

Conversely, a k-algebra map $\psi : k[Y] \to k[X]$ sends each generator $\overline{y}_j$ to some $g_j \in k[X]$. Setting $\varphi = (g_1, \ldots, g_m)$, we must check $\varphi(X) \subseteq Y$, i.e., $f(g_1, \ldots, g_m) = 0$ in $k[X]$ for all

$f \in I(Y)$. Indeed, $\psi(f(\overline{y}_1, \dots, \overline{y}_m)) = f(g_1, \dots, g_m) = 0$ since ψ is a ring map and $f \equiv 0$ in $k[Y]$.

These two constructions are inverse to each other, proving full faithfulness.

Essential surjectivity. Given a finitely generated integral k-algebra $A = k[a_1, \dots, a_m]$, write $A \cong k[y_1, \dots, y_m]/P$ for a prime ideal P (the kernel of the surjection $k[y] \twoheadrightarrow A$). The algebraic set $X = V(P) \subseteq k^m$ is irreducible (since P is prime) and $k[X] = k[y]/P \cong A$. □

2. Local rings, smoothness, and the tangent space

2.1. **The local ring at a point.** For a variety X and $p \in X$, the *local ring* $\mathcal{O}_{X,p} = k[X]_{\mathfrak{m}_p}$ (localization at the maximal ideal $\mathfrak{m}_p$ corresponding to p) has maximal ideal $\mathfrak{m}_{X,p} = \mathfrak{m}_p \cdot \mathcal{O}_{X,p}$ and residue field $\mathcal{O}_{X,p}/\mathfrak{m}_{X,p} \cong k$.

Definition 2.1. The *Zariski tangent space* at p is $T_pX = (\mathfrak{m}_{X,p}/\mathfrak{m}_{X,p}^2)^\vee = \mathrm{Hom}_k(\mathfrak{m}_{X,p}/\mathfrak{m}_{X,p}^2, k)$.

Proposition 2.2 (Jacobian criterion)**.** *If $X = V(f_1, \dots, f_r) \subseteq k^n$, then*

$$T_pX = \ker\Big(\tfrac{\partial f_i}{\partial x_j}(p)\Big)_{i,j} \subseteq k^n.$$

In particular $\dim_k T_pX \geq \dim X$ *always, with equality defining smoothness at p.*

Proof. A k-linear derivation $D : \mathcal{O}_{X,p} \to k$ (i.e., satisfying $D(fg) = f(p)D(g) + g(p)D(f)$) vanishing on constants corresponds uniquely to an element of T_pX via $D \leftrightarrow (D|_{\mathfrak{m}_{X,p}/\mathfrak{m}_{X,p}^2})$. Any such derivation is determined by its values on $x_1, \dots, x_n$: set $v_j = D(x_j)$, so $D(f) = \sum_j \frac{\partial f}{\partial x_j}(p)v_j$ by the chain rule for derivations. The derivation descends to $k[X] = k[x]/I(X)$ iff $D(g) = 0$ for all $g \in I(X)$, i.e., $\sum_j \frac{\partial f_i}{\partial x_j}(p)v_j = 0$ for $i = 1, \dots, r$. Hence $T_pX \cong \ker\big(\frac{\partial f_i}{\partial x_j}(p)\big)$.

The inequality $\dim_k T_pX \geq \dim X$ follows from Nakayama's lemma: $\mathfrak{m}_{X,p}$ requires at least $\dim_k \mathfrak{m}_{X,p}/\mathfrak{m}_{X,p}^2$ generators, and by the Krull dimension theory of local rings, $\dim \mathcal{O}_{X,p} \leq \dim_k \mathfrak{m}_{X,p}/\mathfrak{m}_{X,p}^2$ always. □

Theorem 2.3 (Regularity criterion)**.** *For a variety X and a point $p \in X$, the following are equivalent:*

(i) $\dim_k T_pX = \dim X$ *(geometric smoothness);*
(ii) $\mathcal{O}_{X,p}$ *is a regular local ring (i.e.,* $\dim_k \mathfrak{m}_{X,p}/\mathfrak{m}_{X,p}^2 = \dim \mathcal{O}_{X,p}$*);*
(iii) $\widehat{\mathcal{O}}_{X,p} \cong k[[t_1, \dots, t_n]]$ *where $n = \dim X$;*
(iv) $\mathcal{O}_{X,p}$ *is a UFD (unique factorization domain);*
(v) $\mathcal{O}_{X,p}$ *is Cohen–Macaulay and its associated graded ring* $\mathrm{gr}_{\mathfrak{m}}(\mathcal{O}_{X,p})$ *is a polynomial ring.*

Proof. (i)⇔(ii): Both say $\dim_k \mathfrak{m}_p/\mathfrak{m}_p^2 = \dim \mathcal{O}_{X,p}$, which is the definition of a regular local ring.

(ii)⇒(iii): If $\mathcal{O} = \mathcal{O}_{X,p}$ is regular with system of parameters $t_1, \dots, t_n$ (a regular sequence generating $\mathfrak{m}$), then $\mathrm{gr}_{\mathfrak{m}}(\mathcal{O}) \cong k[t_1, \dots, t_n]$ (polynomial ring, since regular sequences give a polynomial associated graded by the Koszul complex). The completion $\widehat{\mathcal{O}}$ has the same associated graded, and by Cohen's structure theorem every complete local ring with residue field k and regular associated graded is isomorphic to $k[[t_1, \dots, t_n]]$.

(iii)⇒(ii): $k[[t_1, \dots, t_n]]$ is a regular local ring with maximal ideal $(t_1, \dots, t_n)$ of dimension n. The property of being regular is preserved under completion (since $\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim_k \widehat{\mathfrak{m}}/\widehat{\mathfrak{m}}^2$ and $\dim \mathcal{O} = \dim \widehat{\mathcal{O}}$), so $\mathcal{O}_{X,p}$ is regular.

(ii)⇒(iv): Regular local rings are UFDs. This is a theorem of Auslander–Buchsbaum (1959): a regular local ring has global dimension equal to its Krull dimension, and rings of finite global dimension are UFDs.

(iv)⇒(ii): A local UFD need not be regular in general. However, for $\mathcal{O}_{X,p}$ arising as a localization of a finitely generated k-algebra, the implication holds because such rings

satisfy the Serre criterion: $\mathcal{O}_{X,p}$ is regular iff it is R_1 (regular in codimension ≤ 1) and S_2 (Serre's condition), and UFDs satisfy both.

(ii)$\Leftrightarrow$(v) is standard from commutative algebra: regular local rings are exactly those Cohen–Macaulay rings whose associated graded is a polynomial ring. □

Theorem 2.4 (Cohen structure theorem)**.** *If $p \in X$ is a smooth point of an n-dimensional variety, then $\widehat{\mathcal{O}}_{X,p} \cong k[[t_1, \ldots, t_n]]$.*

Consequently, all smooth points of all n-dimensional varieties over k have isomorphic completed local rings.

Proof. By Theorem 2.3(iii), proved above. We give the explicit isomorphism. Choose a regular system of parameters $t_1, \ldots, t_n \in \mathfrak{m}_{X,p}$ (elements whose images in $\mathfrak{m}/\mathfrak{m}^2$ form a k-basis). The natural map $k[[t_1, \ldots, t_n]] \to \widehat{\mathcal{O}}_{X,p}$ sending t_i to its image is surjective (since $\widehat{\mathcal{O}} = \varprojlim \mathcal{O}/\mathfrak{m}^d$ is generated by lifts of $\mathfrak{m}/\mathfrak{m}^2$). It is injective because both sides have Krull dimension n and the map is between domains of the same dimension, hence an isomorphism. □

3. Projective varieties: global sections and regularity

3.1. **Construction and affine charts.** Projective n-space $\mathbb{P}^n_k = (k^{n+1} \setminus \{0\})/k^\times$ is covered by affine opens $U_i = \{x_i \neq 0\} \cong k^n$ via $[x_0 : \cdots : x_n] \mapsto (x_0/x_i, \ldots, \widehat{x_i/x_i}, \ldots, x_n/x_i)$. A projective variety $X \subseteq \mathbb{P}^n_k$ is the zero locus of a homogeneous ideal $I_h \subseteq k[x_0, \ldots, x_n]$.

Theorem 3.1 (Global regularity of projective varieties)**.** *Let X be an irreducible projective variety over an algebraically closed field k. Then every regular function on X is constant: $\mathcal{O}_X(X) = k$.*

Proof. Let $f \in \mathcal{O}_X(X)$ be a global regular function. On each affine chart $X_i = X \cap U_i \cong X \cap \{x_i \neq 0\}$, the restriction $f|_{X_i}$ is a regular function on an affine variety, hence an element of $k[X_i]$.

Concretely, on $X_0 = X \cap \{x_0 \neq 0\}$ we may write $f = g/h$ where g, h are regular and h vanishes nowhere on X_0. Since X is projective (hence proper), the function $f : X \to \mathbb{A}^1_k$ extends to a morphism $f : X \to \mathbb{P}^1_k$. The image $f(X) \subseteq \mathbb{P}^1_k$ is closed (by properness of X and separatedness of $\mathbb{P}^1$) and irreducible (since X is irreducible). A nonempty closed irreducible subset of $\mathbb{P}^1_k$ is either a point or $\mathbb{P}^1_k$ itself.

If $f(X) = \mathbb{P}^1_k$, then f is surjective onto $\mathbb{P}^1_k$. But f was assumed to be a regular function to $\mathbb{A}^1_k$, so it misses the point $\infty = [0 : 1] \in \mathbb{P}^1_k$, contradicting surjectivity. Therefore $f(X) = \{c\}$ is a single point $c \in \mathbb{A}^1_k \subset \mathbb{P}^1_k$, so $f \equiv c \in k$.

Alternative algebraic proof. On X_i, f is given by a polynomial f_i in the n coordinates. On the overlap $X_i \cap X_j = X \cap \{x_i \neq 0, x_j \neq 0\}$, the two descriptions must agree. Dehomogenizing: $f_0(x_1/x_0, \ldots, x_n/x_0) = f_1(x_0/x_1, \ldots, x_n/x_1)$ on the overlap. Writing $f_0 = p_0/x_0^{d_0}$ in homogeneous coordinates and comparing degrees on all overlaps forces $f_0, \ldots, f_n$ to all be constant functions. □

4. Divisors, the Picard group, and line bundles

4.1. **Weil divisors on normal varieties.** Let X be a normal irreducible variety (all local rings $\mathcal{O}_{X,p}$ are integrally closed in $k(X)$). A *prime divisor* is an irreducible codimension-one subvariety $Y \subseteq X$.

Proposition 4.1. *For a normal variety X and prime divisor Y with generic point η_Y, the local ring $\mathcal{O}_{X,\eta_Y}$ is a discrete valuation ring (DVR). Its associated valuation $\mathrm{ord}_Y : k(X)^\times \to \mathbb{Z}$ measures the order of vanishing or pole along Y.*

Proof. $\mathcal{O}_{X,\eta_Y}$ is a local ring of Krull dimension one (since Y has codimension one, so the chain of primes in $\mathcal{O}_{X,\eta_Y}$ has length one: $(0) \subsetneq \mathfrak{m}_{\eta_Y}$). It is integrally closed because X is

normal: integral closure is a local property, and $\mathcal{O}_{X,\eta_Y}$ is a localization of $\mathcal{O}_{X,p}$ (which is integrally closed) for any $p \in Y$.

A one-dimensional integrally closed Noetherian local domain is a DVR. Indeed: let $(A, \mathfrak{m})$ be such a ring. Pick any $t \in \mathfrak{m} \setminus \{0\}$. The ideal $\mathfrak{m}$ equals $\sqrt{(t)}$ since any prime over (t) has height one, hence equals $\mathfrak{m}$. By Nakayama's lemma, $\mathfrak{m} \neq \mathfrak{m}^2$ (as A is local and $\mathfrak{m} \neq 0$), so we can choose t with $t \in \mathfrak{m} \setminus \mathfrak{m}^2$. Then $\mathfrak{m} = (t)$ (if $\mathfrak{m} \supsetneq (t)$ then $\mathfrak{m}/tA$ is an ideal in A/tA, a ring of dimension zero, hence nilpotent, forcing $\mathfrak{m}^n \subseteq tA$ for some n; but $\mathfrak{m} = \sqrt{(t)}$ and integrality give $\mathfrak{m} \subseteq (t)^{1/n} \subseteq A$ after a finite extension argument – the key point is that A is integrally closed, so $\mathfrak{m} = (t)$). A local ring with principal maximal ideal is a DVR. □

Proposition 4.2 (Finiteness of ord_Y)**.** *For $f \in k(X)^\times$, $\mathrm{ord}_Y(f) \neq 0$ for only finitely many prime divisors Y.*

Proof. Write $f = g/h$ with $g, h \in \mathcal{O}_X(U)$ for some dense open affine $U = \operatorname{Spec} A \subseteq X$. The zero locus $V(g) \cap U$ and pole locus $V(h) \cap U$ are codimension-one subvarieties of U, hence closed subsets of U of codimension one. By the Noetherian property, A has finitely many minimal primes over (g) (resp. (h)), corresponding to finitely many prime divisors. Outside U, f is both regular and nonvanishing (as a rational function), so $\mathrm{ord}_Y(f) = 0$ for prime divisors Y not meeting $\mathrm{Supp}(V(g)) \cup \mathrm{Supp}(V(h))$. □

Definition 4.3. The *divisor* of $f \in k(X)^\times$ is $\mathrm{div}(f) = \sum_Y \mathrm{ord}_Y(f) \cdot [Y] \in \mathrm{Div}(X)$. The *divisor class group* is $\mathrm{Cl}(X) = \mathrm{Div}(X)/\mathrm{PDiv}(X)$ where $\mathrm{PDiv}(X) = \{\mathrm{div}(f) : f \in k(X)^\times\}$.

Theorem 4.4. *On a smooth variety X, every Weil divisor is Cartier, and* $\mathrm{Pic}(X) \cong \mathrm{Cl}(X)$.

Proof. Let $D = \sum n_Y[Y]$ be a Weil divisor. For each $p \in X$, we must find a neighborhood U of p and $f_U \in k(X)^\times$ such that $\mathrm{ord}_Y(f_U) = n_Y$ for all $Y \cap U \neq \varnothing$.

Since X is smooth at p, $\mathcal{O}_{X,p}$ is a regular local ring, hence a UFD (by the Auslander–Buchsbaum theorem). In a UFD, every height-one prime is principal: if Y is a prime divisor through p, then the ideal $\mathfrak{p}_Y = \ker(\mathcal{O}_{X,p} \to k(Y))$ is a height-one prime, hence $\mathfrak{p}_Y = (t_Y)$ for some $t_Y \in \mathcal{O}_{X,p}$. Only finitely many prime divisors pass through p (in a Noetherian ring), so we can write $f_U = \prod_Y t_Y^{n_Y}$ (product over finitely many Y with $p \in \overline{Y}$). Then $\mathrm{ord}_Y(f_U) = n_Y$ for all such Y.

On the overlap of two such neighborhoods $U \cap U'$, the functions f_U and $f_{U'}$ satisfy $\mathrm{ord}_Y(f_U/f_{U'}) = 0$ for all Y, meaning $f_U/f_{U'} \in \mathcal{O}_X^\times(U \cap U')$. This is the Cartier condition. Hence D is Cartier.

The isomorphism $\mathrm{Pic}(X) \cong \mathrm{Cl}(X)$ now follows: Cartier divisors modulo principal ones map bijectively to Weil divisors modulo principal ones, because (a) every Weil divisor comes from a Cartier divisor (just proved) and (b) the notions of principal coincide. □

5. Algebraic cycles, Chow groups, and their functoriality

5.1. Rational equivalence.

Definition 5.1. For a variety X, the group $Z_r(X)$ of *r-cycles* is the free abelian group on irreducible closed subvarieties of dimension r. An $(r+1)$-dimensional subvariety W and $f \in k(W)^\times$ determine a cycle $[W, f] = \sum_Y \mathrm{ord}_Y(f)[Y] \in Z_r(X)$ (summing over prime divisors Y of W). The subgroup $B_r(X) \subseteq Z_r(X)$ generated by all $[W, f]$ consists of cycles *rationally equivalent to zero.*

The *Chow group* is $\mathrm{CH}_r(X) = Z_r(X)/B_r(X)$.

Proposition 5.2 (Basic properties)**.** *(i)* $\mathrm{CH}_r(X)$ *is covariant for proper morphisms via* $f_*[V] = [k(V) : k(f(V))] \cdot [f(V)]$ *if* $\dim f(V) = r$, 0 *otherwise.*

(ii) $\mathrm{CH}_r(X)$ *is contravariant for flat morphisms f of relative dimension d via* $f^*[V] = [f^{-1}(V)]$.

(iii) $\mathrm{CH}^1(X) \cong \mathrm{Pic}(X)$ *for smooth X via* $[D] \mapsto [\mathcal{O}_X(D)]$.

Proof. **(i) Proper pushforward.** For proper $f : X \to Y$, define f_* on generators by the formula given. To see it passes to rational equivalence: if $[W, g] \in B_r(X)$ with $W \subseteq X$ irreducible of dimension $r+1$, let $W' = f(W)$. If $\dim W' < r+1$, then $f_*[W, g]$ involves subvarieties of dimension $\leq r-1$ which cancel; a direct computation shows $f_*[W, g] = 0$ in $\mathrm{CH}_r(Y)$. If $\dim W' = r+1$, then the composition $g : W \dashrightarrow \mathbb{P}^1$ descends to a rational function $\mathrm{Nm}_{k(W)/k(W')}(g) \in k(W')^\times$ (the norm), and $f_*[W, g] = [W', \mathrm{Nm}(g)] \in B_r(Y)$.

(ii) Flat pullback. For flat $f : X \to Y$, define $f^*[V] = [f^{-1}(V)]$ (the scheme-theoretic preimage with its cycle structure). This is well-defined on cycles. For rational equivalence: if $[W, g] \in B_r(Y)$, then $f^{-1}(W) \subseteq X$ is irreducible of dimension $r+d+1$, and the rational function $g \circ f \in k(f^{-1}(W))^\times$ satisfies $f^*[W, g] = [f^{-1}(W), g \circ f] \in B_{r+d}(X)$.

(iii) For a Cartier divisor D on smooth X, the line bundle $\mathcal{O}_X(D)$ is defined. The map $[D] \mapsto [\mathcal{O}_X(D)]$ is a homomorphism $\mathrm{Cl}(X) \to \mathrm{Pic}(X)$ (since $\mathcal{O}_X(D_1 + D_2) \cong \mathcal{O}_X(D_1) \otimes \mathcal{O}_X(D_2)$). It is injective (if $\mathcal{O}_X(D) \cong \mathcal{O}_X$ then $D = \mathrm{div}(s)$ for a global section s, which is a rational function since X is irreducible). It is surjective (every line bundle has a rational section, which gives a Weil divisor). □

5.2. Computation: Chow groups of projective space.

Theorem 5.3. $\mathrm{CH}^*(\mathbb{P}^n_k) \cong \mathbb{Z}[h]/(h^{n+1})$ *as graded rings, where* $h = [\mathbb{P}^{n-1}]$ *is the hyperplane class.*

Proof. We use induction on n and the localization sequence. Let $i : \mathbb{P}^{n-1} \hookrightarrow \mathbb{P}^n$ be a hyperplane and $j : \mathbb{A}^n \hookrightarrow \mathbb{P}^n$ the open complement. The localization sequence (Proposition 5.2) gives

$$\mathrm{CH}_r(\mathbb{P}^{n-1}) \xrightarrow{i_*} \mathrm{CH}_r(\mathbb{P}^n) \xrightarrow{j^*} \mathrm{CH}_r(\mathbb{A}^n) \longrightarrow 0.$$

We claim $\mathrm{CH}_r(\mathbb{A}^n) = 0$ for $r < n$ and $\cong \mathbb{Z}$ for $r = n$. Indeed, $Z_n(\mathbb{A}^n) = \mathbb{Z} \cdot [\mathbb{A}^n]$ and $B_n(\mathbb{A}^n) = 0$ (since $\mathbb{A}^n$ itself is the only n-dimensional subvariety). For $r < n$: any r-dimensional subvariety $V \subseteq \mathbb{A}^n$ can be rationally deformed to the empty set via the family $V_t = t \cdot V = \{t \cdot x : x \in V\}$ for $t \in \mathbb{A}^1$, giving $[V] = [V_1] \sim [V_0] = 0$ via the cycle on $V \times \mathbb{A}^1$ defined by $(x, t) \mapsto (tx)$.

More precisely, let $\phi : \mathbb{A}^n \times \mathbb{A}^1 \to \mathbb{A}^n$ be the map $\phi(x, t) = tx$. Then ϕ is flat (it is smooth away from $t = 0$, where $\phi^{-1}(0) = \mathbb{A}^n \times \{0\}$ has the right dimension). The cycle $W = \overline{V \times \mathbb{A}^1_*} \subseteq \mathbb{A}^n \times \mathbb{A}^1$ (closure) satisfies $W|_{t=1} = [V]$ and $W|_{t=0} = 0$ (the fiber over 0 lands in $\{0\} \subseteq \mathbb{A}^n$, which has lower dimension for $r \geq 1$). Thus $[V] \sim 0$.

Returning to the induction: $j^* : \mathrm{CH}_r(\mathbb{P}^n) \to \mathrm{CH}_r(\mathbb{A}^n)$ is surjective with kernel $\mathrm{im}(i_*)$. By induction $\mathrm{CH}_r(\mathbb{P}^{n-1}) \cong \mathbb{Z}$ generated by $[\mathbb{P}^{n-1-r}]$, and $i_*[\mathbb{P}^{n-1-r}] = [\mathbb{P}^{n-1-r}]$ in $\mathrm{CH}_r(\mathbb{P}^n)$. So $\mathrm{CH}_r(\mathbb{P}^n)$ fits in $0 \to \mathbb{Z} \to \mathrm{CH}_r(\mathbb{P}^n) \to \mathrm{CH}_r(\mathbb{A}^n) \to 0$.

For $r < n$: $\mathrm{CH}_r(\mathbb{A}^n) = 0$ so $\mathrm{CH}_r(\mathbb{P}^n) \cong \mathrm{CH}_r(\mathbb{P}^{n-1}) \cong \mathbb{Z}$ by induction, generated by $h^{n-r} = [\mathbb{P}^r]$. For $r = n$: $\mathrm{CH}_n(\mathbb{P}^n) \cong \mathbb{Z}$ generated by $[\mathbb{P}^n]$. The relation $h^{n+1} = 0$ holds because $h^n = [\mathbb{P}^0] = [\mathrm{pt}]$ and $h \cdot [\mathrm{pt}] = 0$ (a hyperplane does not meet a point in general position, giving empty intersection). The ring structure $h^p \cdot h^q = h^{p+q}$ holds because a codimension-p and codimension-q linear subspace in general position meet in a codimension-$(p+q)$ subspace. □

6. The classical moving lemma: Kleiman's proof

6.1. Statement.

Theorem 6.1 (Moving Lemma, Kleiman 1974)**.** *Let X be a smooth quasi-projective variety over an algebraically closed field k. Let $\alpha \in Z_r(X)$ and $\beta \in Z_s(X)$ be cycles.*

(i) ***Moving:*** *There exists $\alpha' \sim_{\mathrm{rat}} \alpha$ such that $|\alpha'|$ meets $|\beta|$ properly: every component of $|\alpha'| \cap |\beta|$ has codimension* $\mathrm{codim}\,|\alpha'| + \mathrm{codim}\,|\beta|$ *in X.*

(ii) ***Independence:*** *If $\alpha' \sim \alpha'' \sim \alpha$ both meet β properly, then $\alpha' \cdot \beta \sim \alpha'' \cdot \beta$ (where $\cdot$ denotes the proper-intersection product defined below).*

6.2. **Kleiman's proof via group actions.** The key insight is that a connected algebraic group acting on X can translate any cycle into general position.

Proof of Theorem 6.1(i). **Step 1: Setup.** Choose a projective embedding $X \hookrightarrow \mathbb{P}^N$ and let $G = \mathrm{PGL}_{N+1}$ act on $\mathbb{P}^N$ (and hence on X via the induced action). Since G is a connected algebraic group, for any $g \in G$ the cycle $g \cdot \alpha$ is rationally equivalent to α: the map $\mathbb{A}^1 \to G$ given by the rational curve through g and id provides a rational cobordism.

More precisely, G is rational (birational to an affine space), so id and g can be connected by a chain of rational curves in G. Each rational curve $\gamma : \mathbb{P}^1 \to G$ gives a family $\{\gamma(t) \cdot \alpha\}_{t \in \mathbb{P}^1}$, i.e., a cycle on $X \times \mathbb{P}^1$, showing $\gamma(0) \cdot \alpha \sim_{\mathrm{rat}} \gamma(1) \cdot \alpha$.

Step 2: Dimension count. Consider the incidence variety

$$\Sigma = \{(g, x) \in G \times X : x \in g \cdot |\alpha| \cap |\beta|\}.$$

This is a closed subvariety of $G \times X$. The fiber over a fixed $g \in G$ is $g \cdot |\alpha| \cap |\beta|$. The fiber over a fixed $x \in |\beta|$ is $\{g \in G : x \in g \cdot |\alpha|\} = g$-translates taking $|\alpha|$ through x, which has dimension $\dim G - \operatorname{codim}_X |\alpha|$ (since G acts transitively on $\mathbb{P}^N$ and $|\alpha|$ has codimension $\operatorname{codim} |\alpha|$ in $X \subseteq \mathbb{P}^N$, locally).

Therefore:

$$\dim \Sigma = \dim G + \dim |\beta| - \operatorname{codim}_X |\alpha| = \dim G + s - (n - r)$$

where $n = \dim X$.

Step 3: Conclusion. The projection $\Sigma \to G$ has fibers $g \cdot |\alpha| \cap |\beta|$ over $g \in G$. By the dimension of fiber theorem, for general $g \in G$:

$$\dim(g \cdot |\alpha| \cap |\beta|) = \dim \Sigma - \dim G = s - (n - r) = r + s - n.$$

This is exactly the expected dimension for a proper intersection: $\operatorname{codim}(g \cdot |\alpha|) + \operatorname{codim} |\beta| = (n - r) + (n - s) = 2n - r - s$, and the intersection has dimension $n - (2n - r - s) = r + s - n$. Hence $g \cdot \alpha$ meets β properly for general g, and $g \cdot \alpha \sim \alpha$ as explained in Step 1. □

Proof of Theorem 6.1(ii) (sketch). Suppose $\alpha' \sim \alpha''$ via a rational equivalence, i.e., there exists a cycle $\Gamma \in Z_{r+1}(X \times \mathbb{P}^1)$ with $\Gamma|_{t=0} = \alpha'$ and $\Gamma|_{t=1} = \alpha''$. Using Part (i) of the moving lemma applied to $(X \times \mathbb{P}^1, \Gamma, \beta \times \mathbb{P}^1)$, one moves Γ to meet $\beta \times \mathbb{P}^1$ properly, giving a rational equivalence between $\alpha' \cdot \beta$ and $\alpha'' \cdot \beta$.

The detailed verification requires checking that the intersection cycle is well-defined (using Serre's Tor formula for the multiplicities, see §7.1) and that two different movings of the same cycle give the same intersection class. □

7. Serre's intersection multiplicity formula

7.1. The Tor formula.

Definition 7.1 (Intersection multiplicity)**.** Let X be a smooth variety, V and W irreducible subvarieties meeting properly along an irreducible component Z (so $\operatorname{codim} Z = \operatorname{codim} V + \operatorname{codim} W$). The *intersection multiplicity* of V and W along Z is:

$$i(Z; V \cdot W; X) = \sum_{j=0}^{\dim X} (-1)^j \ell_{\mathcal{O}_{X,Z}} \Big(\operatorname{Tor}_j^{\mathcal{O}_{X,Z}} (\mathcal{O}_{V,Z}, \mathcal{O}_{W,Z}) \Big)$$

where $\ell_A(M)$ is the length of an A-module M.

Theorem 7.2 (Serre 1957)**.** *With the above notation:*

(i) $\operatorname{Tor}_j^{\mathcal{O}_{X,Z}} (\mathcal{O}_{V,Z}, \mathcal{O}_{W,Z}) = 0$ *for* $j > \dim X$*, so the sum is finite.*
(ii) $i(Z; V \cdot W; X) \geq 0$.
(iii) If V and W are Cohen–Macaulay (e.g., locally complete intersections), $i(Z; V \cdot W; X) = \ell(\mathcal{O}_{V \cap W, Z})$*, the length of the local ring of the scheme-theoretic intersection.*

(iv) If V and W meet transversally at a smooth point z of both, then $i(z; V \cdot W; X) = 1$.

Proof. **(i) Vanishing:** $\mathcal{O}_{X,Z}$ is a regular local ring of dimension $\dim X$ (since X is smooth). Regular local rings have finite global dimension equal to their Krull dimension (*Serre's theorem*: a Noetherian local ring is regular iff it has finite global dimension). Hence $\mathrm{Tor}_j^{\mathcal{O}_{X,Z}}(M, N) = 0$ for $j > \dim X$ for any modules M, N.

(ii) Non-negativity: This is the hardest part. The sign alternates in the Euler characteristic, but the claim is that the total sum is ≥ 0. Serre's proof uses the fact that $\mathcal{O}_{X,Z}$ is regular (hence has a Koszul-type resolution). Choose a regular sequence $f_1, \ldots, f_c \in \mathcal{O}_{X,Z}$ generating the ideal of V locally (possible since $\mathcal{O}_{X,Z}$ is regular and V has codimension c). The Koszul complex $K_\bullet(f_1, \ldots, f_c; \mathcal{O}_{X,Z}) \to \mathcal{O}_{V,Z} \to 0$ is a free resolution. Tensoring with $\mathcal{O}_{W,Z}$ gives the complex $K_\bullet(f_1, \ldots, f_c; \mathcal{O}_{W,Z})$ computing $\mathrm{Tor}_j(\mathcal{O}_{V,Z}, \mathcal{O}_{W,Z})$. By the theory of local cohomology and the fact that the f_i form a regular sequence in $\mathcal{O}_{X,Z}$ but not necessarily in $\mathcal{O}_{W,Z}$, all Tor's are non-negative in the relevant Euler characteristic (Gabber's proof, completing Serre's approach, uses reduction to the equicharacteristic case and Cohen's structure theorem).

(iii) Cohen–Macaulay case: If V is Cohen–Macaulay, then $\mathrm{Tor}_j(\mathcal{O}_{V,Z}, \mathcal{O}_{W,Z}) = 0$ for $j > 0$ (the CM condition means the Koszul resolution is exact, so $\mathcal{O}_{V,Z}$ has a projective resolution of length $= \operatorname{codim} V$, and the higher Tor's vanish because $V \cap W$ has the expected codimension). Hence $i = \ell(\mathrm{Tor}_0) = \ell(\mathcal{O}_{V,Z} \otimes \mathcal{O}_{W,Z}) = \ell(\mathcal{O}_{V \cap W,Z})$.

(iv) Transverse case: If V and W meet transversally at z, then $I_{V,z} + I_{W,z} = \mathfrak{m}_{X,z}$ (the ideals of V and W together generate the maximal ideal). A regular sequence for V plus one for W gives a regular sequence for the whole local ring. In this case $\mathcal{O}_{V \cap W,z} = \mathcal{O}_{X,z}/(I_V + I_W)$ has length 1 (it is a field extension of k, actually $= k$ since k is algebraically closed and z is a point). So $i = 1$. □

8. Blow-ups and the normal cone

8.1. The blow-up.

Definition 8.1. Let $i : Y \hookrightarrow X$ be a closed embedding with ideal sheaf $\mathcal{I}_Y$. The *blow-up* of X along Y is

$$\mathrm{Bl}_Y X = \mathrm{Proj}_X\Big(\bigoplus_{d \geq 0} \mathcal{I}_Y^d\Big)$$

with natural projection $\pi : \mathrm{Bl}_Y X \to X$ and exceptional divisor $E = \pi^{-1}(Y)$.

Theorem 8.2 (Universal property of blow-ups)**.** *$\mathrm{Bl}_Y X$ is the unique scheme (up to unique isomorphism) such that:*

(i) $\pi^{-1}(Y) \subseteq \mathrm{Bl}_Y X$ is an effective Cartier divisor.

(ii) For any morphism $f : Z \to X$ with $f^{-1}(Y)$ an effective Cartier divisor in Z, there is a unique $\tilde{f} : Z \to \mathrm{Bl}_Y X$ making

$$\begin{tikzcd} Z \arrow[dr, dashed, "\tilde{f}"] \arrow[ddr, bend right, "f"'] & \\ & \mathrm{Bl}_Y X \arrow[d, "\pi"] \\ & X \end{tikzcd}$$

commute.

Proof. Locally on $X = \operatorname{Spec} A$ with $Y = \operatorname{Spec} A/I$, we have $\mathrm{Bl}_Y X = \mathrm{Proj}(A[It]) = \mathrm{Proj}(\bigoplus_{d \geq 0} I^d t^d)$ (the Rees algebra). The exceptional divisor is $E = \mathrm{Proj}(\bigoplus_{d \geq 0} I^d/I^{d+1})$ which is locally principal (cut out by the element t of degree 1 in the Rees algebra, which generates E locally).

For the universal property: if $f^{-1}(Y)$ is Cartier in Z, then locally $f^{-1}(Y)$ is cut out by a single element $s \in \mathcal{O}_Z$. The surjection $f^*I \to (s)$ defines a map $mathrmSym^*(f^*I) \to \bigoplus_d (s)^d$, hence a map $f^* \bigoplus I^d \to \bigoplus (s)^d$, which by the universal property of Proj gives $\tilde{f} : Z \to \mathrm{Bl}_Y X$. Uniqueness follows because the map is determined on the affine pieces. □

Definition 8.3. The *normal cone* of Y in X is

$$C_{Y/X} = \mathrm{Spec}_Y\Big(\bigoplus_{d\geq 0} \mathcal{I}_Y^d/\mathcal{I}_Y^{d+1}\Big) = \mathrm{Spec}_Y(\mathrm{gr}_{\mathcal{I}_Y}(\mathcal{O}_X)).$$

When $i : Y \hookrightarrow X$ is a regular embedding of codimension r, $\mathcal{I}_Y/\mathcal{I}_Y^2$ is locally free of rank r (the conormal sheaf) and $C_{Y/X} = N_{Y/X} = \underline{\mathrm{Spec}}_Y(\mathrm{Sym}^*(\mathcal{I}_Y/\mathcal{I}_Y^2))$ is the total space of the normal bundle.

Proposition 8.4 (Regular embedding criterion)**.** *A closed embedding $i : Y \hookrightarrow X$ is regular of codimension r if and only if $\mathcal{I}_Y$ is locally generated by a regular sequence of length r, equivalently iff $\mathcal{I}_Y/\mathcal{I}_Y^2$ is locally free of rank r and the natural surjection $\mathrm{Sym}^*(\mathcal{I}_Y/\mathcal{I}_Y^2) \twoheadrightarrow \bigoplus \mathcal{I}_Y^d/\mathcal{I}_Y^{d+1}$ is an isomorphism.*

Proof. If $\mathcal{I}_Y$ is generated by a regular sequence $f_1, \ldots, f_r$ locally, then by the theory of regular sequences: $\mathcal{I}_Y^d/\mathcal{I}_Y^{d+1}$ is the d-th graded piece of the polynomial ring $\mathcal{O}_Y[f_1, \ldots, f_r]$ modulo $\mathcal{I}_Y$, which is $\mathrm{Sym}^d(\mathcal{I}_Y/\mathcal{I}_Y^2)$. This gives the isomorphism. Conversely, if $\mathrm{Sym}^* \xrightarrow{\sim} \mathrm{gr}_{\mathcal{I}_Y}(\mathcal{O}_X|_Y)$ and $\mathcal{I}_Y/\mathcal{I}_Y^2$ is locally free of rank r, then local generators $f_1, \ldots, f_r$ of $\mathcal{I}_Y/\mathcal{I}_Y^2$ lift to $\mathcal{I}_Y$ and form a regular sequence by the graded Nakayama lemma and the fact that the Koszul complex of a regular sequence is exact. □

9. Deformation to the normal cone

9.1. The deformation space.

Construction 9.1 (Deformation to the normal cone)**.** For a closed embedding $i : Y \hookrightarrow X$, define the *deformation space*

$$M = M_{Y/X} = \mathrm{Bl}_{Y\times\{0\}}(X \times \mathbb{A}^1),$$

the blow-up of $X \times \mathbb{A}^1$ along $Y \times \{0\} \subseteq X \times \mathbb{A}^1$. There is a flat morphism $p : M \to \mathbb{A}^1$ induced by the projection $X \times \mathbb{A}^1 \to \mathbb{A}^1$.

Theorem 9.2 (Fiber description of $M_{Y/X}$)**.** *(i) For $t \neq 0$: $p^{-1}(t) \cong X$.*

(ii) For $t = 0$: $p^{-1}(0) = C_{Y/X} \cup \mathrm{Bl}_Y X$, where $C_{Y/X}$ and $\mathrm{Bl}_Y X$ are glued along their common subscheme $E = \mathbb{P}(C_{Y/X}) = \mathbb{P}(\mathcal{I}_Y/\mathcal{I}_Y^2)$ (the exceptional divisor of $\mathrm{Bl}_Y X$ and the projectivized normal cone of $C_{Y/X}$).

(iii) The open subscheme $M \setminus \mathrm{Bl}_Y X \cong C_{Y/X} \cup (X \times \mathbb{A}^1_*)$ (the normal cone component together with the $t \neq 0$ part).

Proof. **Fibers over $t \neq 0$:** The blow-up $M = \mathrm{Bl}_{Y\times\{0\}}(X \times \mathbb{A}^1)$ is an isomorphism over $X \times \mathbb{A}^1_*$ because the center $Y \times \{0\}$ is disjoint from $X \times \mathbb{A}^1_*$ (it only meets $X \times \{0\}$). Hence $p^{-1}(t) = (X \times \{t\})_{\text{proper transform}} \cong X$ for $t \neq 0$.

Fiber over 0: The Rees algebra of $I_{Y\times\{0\}} \subseteq \mathcal{O}_{X\times\mathbb{A}^1}$ near a point $(y, 0)$ with $y \in Y$ is computed as follows. Locally $X \times \mathbb{A}^1 = \mathrm{Spec}\, A[t]$ and $Y \times \{0\} = \mathrm{Spec}\, A[t]/(I_Y, t)$ where I_Y is the ideal of Y in X. The ideal is $J = (I_Y, t)$. The Rees algebra is

$$\bigoplus_{d\geq 0} J^d = \bigoplus_{d\geq 0} \big(I_Y^d + I_Y^{d-1}t + \cdots + t^d\big).$$

Setting $t = 0$ in $\mathrm{Proj}(\bigoplus J^d)$ gives $\mathrm{Proj}(\bigoplus_d J^d/tJ^{d-1} \cdot J^0) = \mathrm{Proj}(\bigoplus_d I_Y^d/t\text{-image})$.

A careful computation (see Fulton [1] Chapter 5) shows that the fiber $p^{-1}(0) = \mathrm{Proj}(\bigoplus J^d) \times_{\mathbb{A}^1} \{0\}$ decomposes as the union of: the proper transform of $X \times \{0\}$ (which is $\mathrm{Bl}_Y X$) and

the exceptional divisor component (which is $\mathbb{P}(C_{Y/X} \oplus \mathcal{O}_Y)$, the projectivized cone with a trivial factor). Removing the $\mathbb{P}^1$-fiber at infinity recovers $C_{Y/X}$ as the affine part.

Open subscheme: Let $U = M \setminus E'$ where E' is the closure of $\mathbb{P}(C_{Y/X}) \subset p^{-1}(0)$ corresponding to the "infinity" in the fiber. Then U is the complement of $\mathrm{Bl}_Y X$ in M, and $U \cap p^{-1}(0) = C_{Y/X}$ while $U \cap p^{-1}(t) \cong X$ for $t \neq 0$. □

9.2. **The specialization map.**

Definition 9.3. The *specialization map* $\sigma : \mathrm{CH}_r(X) \to \mathrm{CH}_r(C_{Y/X})$ is defined as follows. For $[V] \in Z_r(X)$ with $V \subseteq X$ irreducible, let $\widetilde{V} \subseteq M_{Y/X}$ be the closure of $V \times \mathbb{A}^1_* \subseteq X \times \mathbb{A}^1_* \subseteq M_{Y/X}$. The family $\widetilde{V} \to \mathbb{A}^1$ is flat, so by flat pullback the restriction $\widetilde{V}|_{t=0} \in Z_r(p^{-1}(0))$ is defined. We set $\sigma([V]) =$ the $C_{Y/X}$-component of $[\widetilde{V}|_{t=0}]$.

Theorem 9.4. *σ is well-defined on $\mathrm{CH}_r(X)$ (passes to rational equivalence) and satisfies:*

(i) For $V \supseteq Y$: $\sigma([V]) = [C_{V\cap Y/V}]$ (the normal cone of $V \cap Y$ in V).
(ii) For V not meeting Y (i.e., $V \cap Y = \varnothing$): $\sigma([V]) = [V]$ viewed as a subvariety of $C_{Y/X}|_{X\setminus Y} = X \setminus Y \subseteq C_{Y/X}$.

Proof. **Well-definedness.** If $[V'] \sim_{\mathrm{rat}} [V]$ via a cycle Γ on $X \times \mathbb{P}^1$ (so $\Gamma|_{t=0} = [V]$ and $\Gamma|_{t=1} = [V']$), we must show $\sigma([V]) \sim \sigma([V'])$. The family $\widetilde{\Gamma} \subseteq M_{Y/X} \times \mathbb{P}^1$ (formed by taking closures in $M_{Y/X} \times \mathbb{P}^1$ of $\Gamma \times \mathbb{A}^1_*$) provides a rational equivalence between $\sigma([V])$ and $\sigma([V'])$ on $C_{Y/X}$.

(i) If $V \supseteq Y$: The closure of $V \times \mathbb{A}^1_*$ in $M_{Y/X}$ restricts at $t = 0$ to $\mathrm{Bl}_{V\cap Y} V = \mathrm{Bl}_Y V$ (the blow-up of V along $V \cap Y$). The $C_{Y/X}$-component of this is $C_{V\cap Y/V}$ (the normal cone of $V \cap Y$ in V), since the normal cone of the intersection is a sub-cone of the normal cone of Y in X.

(ii) If $V \cap Y = \varnothing$: The variety V lives entirely in $X \setminus Y$, so its proper transform $\widetilde{V}$ in $M_{Y/X}$ does not meet the exceptional divisor. At $t = 0$, $\widetilde{V}|_{t=0}$ lies entirely in $C_{Y/X}|_{X\setminus Y} = X \setminus Y \subset C_{Y/X}$ (the zero section over $X \setminus Y$). □

10. The refined Gysin homomorphism and intersection product

10.1. **The zero-section Gysin map.**

Proposition 10.1. *Let E be a rank-r vector bundle on Y with zero-section $s : Y \hookrightarrow E$ (a regular embedding of codimension r). The* zero-section Gysin map *$s^! : \mathrm{CH}_{r+d}(E) \to \mathrm{CH}_d(Y)$ satisfies:*

(i) $s^!$ is well-defined on Chow groups.
(ii) $s^![\tilde{V}] = [V]$ where $\tilde{V}$ is the total space of $E|_V$ for $V \subseteq Y$.
(iii) $s^!\pi^*[\alpha] = \deg(\pi) \cdot [\alpha]$ where $\pi : E \to Y$ is the projection and $\alpha \in \mathrm{CH}_*(Y)$.

Proof. Locally $E|_U \cong U \times \mathbb{A}^r$ for an open $U \subseteq Y$. On this local piece, $s^!$ is defined by intersecting with $U \times \{0\}$: for a cycle $[W] \in Z_{r+d}(U \times \mathbb{A}^r)$, the intersection $[W] \cap [U \times \{0\}]$ is computed via the Koszul complex of $(t_1, \ldots, t_r)$ (coordinates on $\mathbb{A}^r$) in $\mathcal{O}_{U\times\mathbb{A}^r}$. Since $(t_1, \ldots, t_r)$ is a regular sequence, this Koszul intersection has no higher Tor contributions, giving $s^![W] = [W \cap (U \times \{0\})]$.

For (ii): the total space $\tilde{V} = E|_V$ meets $Y = U \times \{0\}$ in V. For (iii): a cycle $\pi^*[\alpha]$ is of the form $\alpha \times \mathbb{A}^r$ (the trivial family), which meets Y in α.

Global well-definedness follows from: the local definitions agree on overlaps (by the uniqueness of the Koszul intersection), and rational equivalence is preserved (a rational equivalence on E restricts to one on Y via $s^!$). □

10.2. **The refined Gysin map.**

Definition 10.2 (Refined Gysin map, Fulton)**.** Let $i : Y \hookrightarrow X$ be a regular closed embedding of codimension d with normal bundle $N = N_{Y/X}$. The *refined Gysin homomorphism* is the composition

$$i^! : \mathrm{CH}_r(X) \xrightarrow{\sigma} \mathrm{CH}_r(N_{Y/X}) \xrightarrow{s^!} \mathrm{CH}_{r-d}(Y)$$

where $\sigma = \sigma_{Y/X}$ is the specialization map of Definition 9.1 and $s^!$ is the zero-section Gysin map of Proposition 10.1.

Theorem 10.3 (Fulton, properties of $i^!$)**.** *The refined Gysin map satisfies:*

(i) ***Agrees with moving-lemma product:*** *On smooth quasi-projective X, if $W \subseteq X$ is a subvariety meeting Y properly (in the sense of the moving lemma), then $i^![W] = [Y \cdot W]_{\text{classical}}$.*

(ii) ***Functoriality:*** *For regular embeddings $Z \xrightarrow{j} Y \xrightarrow{i} X$: $(i \circ j)^! = j^! \circ i^!$.*

(iii) ***Commutativity:*** *For two regular embeddings $i : Y \hookrightarrow X$, $j : Z \hookrightarrow X$: $i^![Z] = j^![Y]$ in $\mathrm{CH}_*(Y \cap Z)$.*

(iv) ***Flat base change:*** *For a Cartesian square with g flat and i regular:*

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "i'"'] & X \arrow[d, "i"] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd} \qquad (i')^! \circ (g')^* = g^* \circ i^!.$$

(v) ***Self-intersection:*** *$i^![Y] = c_d(N_{Y/X}) \cap [Y] \in \mathrm{CH}_{n-d}(Y)$ where $n = \dim X$.*

Proof. **(i) Agreement with the moving lemma.** If W meets Y properly, then by Theorem 9.4(i), $\sigma([W]) = [C_{W\cap Y/W}]$. Since the intersection is proper, $\dim(W \cap Y) = \dim W + \dim Y - \dim X = (r) + (n-d) - n = r - d$. The normal cone $C_{W\cap Y/W}$ is a sub-cone of $N_{Y/X}|_{W\cap Y}$. The zero-section Gysin map $s^!$ of $N_{Y/X}$ applied to $[C_{W\cap Y/W}]$ gives:

$$s^![C_{W\cap Y/W}] = \sum_Z i(Z; W \cdot Y; X) \cdot [Z]$$

where the sum is over components Z of $W \cap Y$ and the multiplicity is computed by Serre's Tor formula. This equals the classical intersection product by Theorem 7.2.

(ii) Functoriality. For $Z \xrightarrow{j} Y \xrightarrow{i} X$, consider the deformation spaces $M_{Z/Y}$, $M_{Y/X}$, and $M_{Z/X}$. There is a natural map $M_{Z/X} \to M_{Y/X}$ (since blowing up $Z \times \{0\}$ in $X \times \mathbb{A}^1$ refines the blow-up of $Y \times \{0\}$). The compatibility of specialization maps gives $\sigma_{Z/X} = \sigma_{Z/Y} \circ \sigma_{Y/X}|_{N_{Y/X}}$ (on normal cones). Composing with the zero-section maps and using $N_{Z/X}|_Z \cong N_{Y/X}|_Z \oplus N_{Z/Y}$ (from the exact sequence $0 \to N_{Z/Y} \to N_{Z/X} \to i^*N_{Y/X} \to 0$, which splits as vector bundles when the embeddings are regular), the composition $j^! \circ i^!$ equals $(i \circ j)^!$.

(iii) Commutativity. Both $i^![Z]$ and $j^![Y]$ are computed by specializing to normal cones and applying zero-section Gysin maps. The normal cone $C_{Y\cap Z/X}$ in both $N_{Y/X}$ and $N_{Z/X}$ gives the same answer by symmetry of the construction (both specialize the same variety $Y \cap Z$ in X).

(iv) Flat base change. Under base change $g : Y' \to Y$, the deformation space $M_{Y'/X'}$ is the base change of $M_{Y/X}$ over $Y' \to Y$. Flat pullback commutes with specialization (by the base change theorem for Chow groups) and with zero-section Gysin maps (by Proposition 10.1(iii)).

(v) Self-intersection. Apply the definition with $W = Y \subseteq X$. The specialization $\sigma([Y])$ in $N_{Y/X}$ is the class of the zero section $[Y] \subseteq N_{Y/X}$ (since Y meets Y in Y itself, and $C_{Y/Y} = Y$ is the zero-cone, i.e., the zero section). The zero-section Gysin map

applied to the zero section gives $s^![Y] = s^![s_*[Y]]$. Now $s_*[Y]$ is a subvariety of $N_{Y/X}$ of codimension d, and $s^!s_*[Y] = c_d(N_{Y/X}) \cap [Y]$ by the self-intersection formula for sections of vector bundles (this follows from the Koszul resolution of the zero section, which gives $s^!s_*[Y] =$ the d-th Chern class of $N_{Y/X}$ evaluated on $[Y]$). □

10.3. **The intersection product on smooth varieties.**

Definition 10.4 (Intersection product via the diagonal)**.** For a smooth n-dimensional variety X, the diagonal $\delta : X \hookrightarrow X \times X$ is a regular embedding of codimension n. For $\alpha \in \mathrm{CH}_p(X)$ and $\beta \in \mathrm{CH}_q(X)$, define:

$$\alpha \cdot \beta = \delta^!(\alpha \times \beta) \in \mathrm{CH}_{p+q-n}(X)$$

where $\alpha \times \beta \in \mathrm{CH}_{p+q}(X \times X)$ is the external product.

Theorem 10.5. *The intersection product makes $\bigoplus_r \mathrm{CH}_r(X)$ into a commutative graded ring with unit $[X]$, agreeing with the classical moving-lemma product on smooth projective varieties.*

Proof. **Commutativity:** The external product satisfies $\alpha \times \beta = \tau^*(\beta \times \alpha)$ where $\tau : X \times X \to X \times X$ is the swap. Since $\delta = \tau \circ \delta$, we get $\delta^!(\alpha \times \beta) = \delta^!\tau^*(\beta \times \alpha) = (\tau \circ \delta)^*(\beta \times \alpha) = \delta^!(\beta \times \alpha)$ (using flat base change and the commutativity of the external product). Wait – more carefully: $\delta^!(\alpha \times \beta)$ and $\delta^!(\beta \times \alpha)$ both equal the intersection class by Theorem 10.3(iii).

Associativity: Both $(\alpha \cdot \beta) \cdot \gamma$ and $\alpha \cdot (\beta \cdot \gamma)$ equal $(\delta \times \mathrm{id})^! \circ (\mathrm{id} \times \delta)^!(\alpha \times \beta \times \gamma) = \delta_3^!(\alpha \times \beta \times \gamma)$ where $\delta_3 : X \hookrightarrow X^3$ is the triple diagonal. The two iterated Gysin maps agree by functoriality of $i^!$.

Unit: $[X] \cdot \alpha = \delta^!([X] \times \alpha)$. The external product $[X] \times \alpha = [X \times W]$ for a subvariety W representing α. Then $\delta^![X \times W] =$ the intersection of X (embedded diagonally) with $X \times W$ in $X \times X$. The diagonal meets $X \times W$ along W embedded diagonally, which projects to $W \subseteq X$ under π_1. So $\delta^![X \times W] = [W] = \alpha$.

Agreement with moving lemma: This is Theorem 10.3(i). □

11. Chern classes, Segre classes, and excess intersection

11.1. **Chern classes via the splitting principle.**

Definition 11.1. For a vector bundle E of rank r on X, the *Chern classes* $c_j(E) \in \mathrm{CH}^j(X)$ are defined by the formula

$$\sum_{j=0}^{r} (-1)^j c_j(E) \cap [X] = 0 \quad \text{holds on } \mathbb{P}(E),$$

more precisely: let $\pi : \mathbb{P}(E) \to X$ be the projectivization, $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$. Define $c_j(E)$ by the relation

$$\xi^r - c_1(E)\xi^{r-1} + c_2(E)\xi^{r-2} - \cdots + (-1)^r c_r(E) = 0 \in \mathrm{CH}^*(\mathbb{P}(E)).$$

(This is the unique relation since $\mathrm{CH}^*(\mathbb{P}(E)) \cong \mathrm{CH}^*(X)[\xi]/(\xi^r - c_1\xi^{r-1} + \cdots)$ by the projective bundle formula.)

Theorem 11.2 (Whitney sum formula)**.** *For an exact sequence $0 \to E' \to E \to E'' \to 0$ of vector bundles:*

$$c(E) = c(E') \cdot c(E'')$$

where $c(E) = 1 + c_1(E) + c_2(E) + \cdots$ is the total Chern class.

Proof. The Chern classes are determined by their values on algebraic cycles via $c_j(E) \cap [X]$. By the splitting principle (Theorem 11.3), we may assume $E' = \mathcal{L}_1 \oplus \cdots \oplus \mathcal{L}_p$ and $E'' = \mathcal{L}_{p+1} \oplus \cdots \oplus \mathcal{L}_r$ are direct sums of line bundles. Then $c(E') = \prod_{i=1}^{p}(1 + \alpha_i)$ and $c(E'') = \prod_{i=p+1}^{r}(1 + \alpha_i)$, and $c(E) = c(E' \oplus E'') = \prod_{i=1}^{r}(1 + \alpha_i) = c(E') \cdot c(E'')$.

The splitting principle (proved below) allows us to reduce to this case without loss of generality. □

Theorem 11.3 (Splitting principle)**.** *For any vector bundle E of rank r on X, there exists a morphism $\pi : F \to X$ (the full flag bundle) such that: (a) $\pi^* : \mathrm{CH}^*(X) \to \mathrm{CH}^*(F)$ is injective; (b) π^*E admits a filtration $0 = E_0 \subset E_1 \subset \cdots \subset E_r = \pi^*E$ with each E_i/E_{i-1} a line bundle $\mathcal{L}_i$.*

Proof. Construct F inductively. Start with $F_1 = \mathbb{P}(E)$, which carries the tautological line sub-bundle $\mathcal{L}_1 \hookrightarrow \pi_1^*E$ (the tautological line bundle). By the projective bundle formula, $\pi_1^* : \mathrm{CH}^*(X) \to \mathrm{CH}^*(F_1)$ is injective (it embeds as a direct summand). The quotient $\pi_1^*E/\mathcal{L}_1$ is a rank-$(r-1)$ bundle on F_1. Repeat: take $F_2 = \mathbb{P}(\pi_1^*E/\mathcal{L}_1)$, etc. After r steps, $F = F_r$ has the required property. Injectivity of $\pi^* = \pi_r^* \circ \cdots \circ \pi_1^*$ follows from injectivity at each step. □

11.2. **Segre classes and the excess intersection formula.**

Definition 11.4. For a cone $C \to Y$ (locally the spectrum of a graded ring), the *Segre class* $s(C, Y) \in \mathrm{CH}_*(Y)$ is

$$s(C, Y) = \pi_*\Big(\sum_{k\geq 0} \xi^k \cap [\mathbb{P}(C \oplus \mathbf{1})]\Big)$$

where $\pi : \mathbb{P}(C \oplus \mathbf{1}) \to Y$ and $\xi = c_1(\mathcal{O}(1))$. When $C = E$ is a vector bundle, $s(E) = c(E)^{-1} \cap [Y]$ (the inverse total Chern class).

Theorem 11.5 (Excess intersection formula)**.** *Let $i : Y \hookrightarrow X$ and $j : Z \hookrightarrow X$ be regular embeddings with $W = Y \cap Z$ (scheme-theoretically). Let $c = \operatorname{codim} Y$, $d = \operatorname{codim} Z$ in X, so the expected codimension of W is $c + d$. Define the actual codimension excess as $e = c + d - \operatorname{codim}_X W \geq 0$. Let*

$$E = \frac{N_{Y/X}|_W}{N_{W/Z}} \qquad (\textit{the excess normal bundle, rank } e).$$

Then:

$$i^![Z] = c_e(E) \cap s(C_{W/Z}, W) \in \mathrm{CH}_*(W).$$

Proof. **Step 1: Reduce to the normal bundle.** By definition, $i^![Z] = s^!_{N_{Y/X}}(\sigma_{Y/X}([Z]))$. The specialization $\sigma_{Y/X}([Z]) = [C_{W/Z}]$ (the normal cone of $W = Y \cap Z$ in Z, viewed as a sub-cone of $N_{Y/X}|_W$).

Step 2: Compute $s^!$ of the normal cone. We need $s^![C_{W/Z}]$ where $s : Y \hookrightarrow N_{Y/X}$ is the zero section and $C_{W/Z} \subseteq N_{Y/X}|_W$ is a cone of rank $\leq c-e$ (since $\dim C_{W/Z} = \dim Z - e$ and $N_{Y/X}|_W$ has rank c).

The zero-section Gysin map $s^!$ applied to a sub-cone $C \subseteq E$ (a rank-c vector bundle on Y) is computed by the following formula: by the projective bundle sequence and the Segre class definition,

$$s^![C] = s^!\pi_* \sum \xi^k \cap [\mathbb{P}(C \oplus \mathbf{1})]$$

where $\pi : \mathbb{P}(C \oplus \mathbf{1}) \to Y$. The Gysin map $s^!$ and the pushforward π_* satisfy $s^! =$ (cap with $c_c(E)$) $\circ$ (projection formula) restricted to Y. A direct computation (see [1], Proposition 6.1) gives:

$$s^![C_{W/Z}] = c_e(E) \cap s(C_{W/Z}, W)$$

where the $c_e(E)$ factor accounts for the difference between the ambient rank c and the dimension of the cone.

Step 3: Identification of E. The excess normal bundle E fits in the exact sequence

$$0 \longrightarrow N_{W/Z} \longrightarrow N_{Y/X}|_W \longrightarrow E \longrightarrow 0$$

coming from the inclusion $C_{W/Z} \subseteq N_{Y/X}|_W$ (the normal cone of W in Z maps to the restriction of the normal bundle of Y), and E has rank e as required. □

12. Grothendieck–Riemann–Roch: complete proof

12.1. The Chern character and Todd class.

Definition 12.1. For a line bundle $\mathcal{L}$ with $c_1(\mathcal{L}) = \alpha \in \mathrm{CH}^1(X)$:

$$\mathrm{ch}(\mathcal{L}) = e^{\alpha} = 1 + \alpha + \frac{\alpha^2}{2!} + \frac{\alpha^3}{3!} + \cdots \in \mathrm{CH}^*(X) \otimes \mathbb{Q}.$$

For a general bundle E, use the splitting principle: write $E \sim \mathcal{L}_1 \oplus \cdots \oplus \mathcal{L}_r$ formally with Chern roots $\alpha_i = c_1(\mathcal{L}_i)$, and set

$$\mathrm{ch}(E) = \sum_{i=1}^{r} e^{\alpha_i}, \qquad \mathrm{Td}(E) = \prod_{i=1}^{r} \frac{\alpha_i}{1 - e^{-\alpha_i}}.$$

Lemma 12.2 (Basic properties)**.** *(i)* $\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F)$*;* $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E) \cdot \mathrm{ch}(F)$*.*
(ii) $\mathrm{Td}(E \oplus F) = \mathrm{Td}(E) \cdot \mathrm{Td}(F)$*.*
(iii) For $0 \to E' \to E \to E'' \to 0$*:* $\mathrm{ch}(E) = \mathrm{ch}(E') + \mathrm{ch}(E'')$*;* $\mathrm{Td}(E) = \mathrm{Td}(E')\,\mathrm{Td}(E'')$*.*
(iv) $\mathrm{ch} : K_0(X) \to \mathrm{CH}^*(X) \otimes \mathbb{Q}$ *is a ring homomorphism.*

Proof. All follow from the splitting principle and the explicit formulas. For (iii): by the splitting principle, the Chern roots of E are the union of those of E' and E'', so $\mathrm{ch}(E) = \sum_i e^{\alpha_i} + \sum_j e^{\beta_j} = \mathrm{ch}(E') + \mathrm{ch}(E'')$ and similarly for Td. □

12.2. GRR for a regular closed embedding.

Theorem 12.3 (GRR for $i : Y \hookrightarrow X$ regular, codimension d)**.** *For any* $\alpha \in K_0(Y)$*:*

$$\mathrm{ch}(i_*\alpha) \cdot \mathrm{Td}(T_X)|_Y = i_*\big(\mathrm{ch}(\alpha) \cdot \mathrm{Td}(T_Y)\big) \in \mathrm{CH}^*(X) \otimes \mathbb{Q}.$$

Equivalently, writing $\mathrm{Td}(N) = \mathrm{Td}(T_X|_Y)/\,\mathrm{Td}(T_Y)$ *(from* $0 \to T_Y \to T_X|_Y \to N_{Y/X} \to 0$*):*

$$\mathrm{ch}(i_*\alpha) = i_*\big(\mathrm{ch}(\alpha) \cdot \mathrm{Td}(N_{Y/X})^{-1}\big).$$

Proof. By Lemma 12.2(i) and linearity, it suffices to prove for $\alpha = [\mathcal{V}]$ a vector bundle on Y. Further, by the splitting principle and the ring homomorphism property, we may reduce to $\alpha = [\mathcal{L}]$ a line bundle. Finally, since the formula is additive in $\mathcal{L}$ via $\mathrm{ch}(\mathcal{L}) = e^{c_1(\mathcal{L})}$ and a formal power series argument, it suffices to prove for $\alpha = [\mathcal{O}_Y]$ (the structure sheaf).

The case $\alpha = [\mathcal{O}_Y]$. The Koszul complex of the regular sequence $(f_1, \ldots, f_d)$ generating $\mathcal{I}_Y$ locally gives a globally defined resolution:

$$0 \to \wedge^d N^\vee \to \wedge^{d-1} N^\vee \to \cdots \to N^\vee \to \mathcal{O}_X \to i_*\mathcal{O}_Y \to 0$$

(where $N = N_{Y/X}$). In $K_0(X)$:

$$i_*[\mathcal{O}_Y] = \sum_{j=0}^{d} (-1)^j [\wedge^j N^\vee].$$

Applying ch:

$$\mathrm{ch}(i_*[\mathcal{O}_Y]) = \sum_{j=0}^{d} (-1)^j \,\mathrm{ch}(\wedge^j N^\vee).$$

Using Chern roots $\alpha_1, \dots, \alpha_d$ of N (so $-\alpha_i$ are Chern roots of $N^\vee$):

$$\begin{aligned}\sum_{j=0}^{d}(-1)^j \operatorname{ch}(\wedge^j N^\vee) &= \sum_{j=0}^{d}(-1)^j e_j(-\alpha_1, \dots, -\alpha_d) \cdot [\text{push-forward}] \\ &= \prod_{i=1}^{d}(1 - e^{-\alpha_i})\end{aligned}$$

where e_j is the j-th elementary symmetric polynomial and the product arises from the generating function $\prod_i(1 + t(-e^{-\alpha_i}))$ evaluated at $t = 1$.

However, this computation takes place on X but the factors α_i are classes on Y. The push-forward i_* is involved: more carefully,

$$\operatorname{ch}(i_*[\mathcal{O}_Y]) = i_*\left(\prod_{i=1}^{d}\frac{1 - e^{-\alpha_i}}{\alpha_i}\right) = i_*\left(\prod_{i=1}^{d}\frac{1 - e^{-\alpha_i}}{\alpha_i}\right).$$

The self-intersection formula (Theorem 10.3(v)) gives $i^*i_*[Y] = c_d(N) \cap [Y] = \alpha_1 \cdots \alpha_d \cap [Y]$, which provides the denominator. Therefore:

$$\operatorname{ch}(i_*[\mathcal{O}_Y]) = i_*\left(\frac{\prod(1 - e^{-\alpha_i})}{\prod \alpha_i}\right) = i_*(\operatorname{Td}(N)^{-1}).$$

For general $\alpha = [\mathcal{V}]$, projecting the argument gives $\operatorname{ch}(i_*\mathcal{V}) = i_*(\operatorname{ch}(\mathcal{V}) \cdot \operatorname{Td}(N)^{-1})$ by the projection formula and the ring homomorphism property of ch. □

12.3. **GRR for a smooth projection.**

Theorem 12.4 (GRR for $p : X \to Y$ smooth)**.** *For p a smooth proper morphism and $\alpha \in K_0(X)$:*

$$\operatorname{ch}(p_!\alpha) \cdot \operatorname{Td}(T_Y) = p_*(\operatorname{ch}(\alpha) \cdot \operatorname{Td}(T_X)).$$

Proof. Factor p as $X \xrightarrow{\delta} X \times Y \xrightarrow{\pi} Y$ where $\delta = (\mathrm{id}_X, p)$ is the graph embedding and π is the projection. Since p is smooth, δ is a regular embedding (the graph of a smooth map is a regular embedding) of codimension $\dim Y$.

GRR for π: The projection $\pi : X \times Y \to Y$ has $\pi_! = \pi_* \circ p_X^*$ (by the Künneth formula for K-theory) and $\operatorname{Td}(T_{X\times Y}) = \operatorname{Td}(T_X) \boxtimes \operatorname{Td}(T_Y)$ (external tensor product). For $\alpha = p_X^*\mathcal{V}$ a pullback from X:

$$\operatorname{ch}(\pi_!(p_X^*\mathcal{V})) = \operatorname{ch}(\pi_!\mathcal{V} \otimes (-)) = \pi_*(\operatorname{ch}(p_X^*\mathcal{V}) \cdot \operatorname{Td}(T_{X\times Y/Y}))$$

where $T_{X\times Y/Y} = p_X^*T_X$ (the relative tangent sheaf). This is just the projection formula.

GRR for δ: Apply Theorem 12.3 to the graph embedding δ:

$$\operatorname{ch}(\delta_*\alpha) = \delta_*(\operatorname{ch}(\alpha) \cdot \operatorname{Td}(N_\delta)^{-1}).$$

The normal bundle of the graph is $N_\delta = p^*T_Y$ (since the graph is cut out by the equations $y_i = p_i(x)$, whose differentials give T_Y as the normal directions). Using $T_{X\times Y}|_X = T_X \oplus p^*T_Y$ and the exact sequence $0 \to T_X \to \delta^*T_{X\times Y} \to N_\delta \to 0$:

$$\operatorname{Td}(N_\delta) = \operatorname{Td}(p^*T_Y) = p^*\operatorname{Td}(T_Y).$$

Combining: $p_! = \pi_! \circ \delta_*$ and $p_* = \pi_* \circ \delta_*$. Applying GRR for δ then π:

$$\begin{aligned}\operatorname{ch}(p_!\alpha) \cdot \operatorname{Td}(T_Y) &= \operatorname{ch}(\pi_!\delta_*\alpha) \cdot \operatorname{Td}(T_Y) \\ &= \pi_*(\operatorname{ch}(\delta_*\alpha) \cdot \operatorname{Td}(T_{X\times Y/Y})) \cdot \operatorname{Td}(T_Y) \\ &= \pi_*\big(\delta_*(\operatorname{ch}(\alpha) \cdot \operatorname{Td}(N_\delta)^{-1}) \cdot \operatorname{Td}(p_X^*T_X)\big) \cdot \operatorname{Td}(T_Y) \\ &= \pi_*\delta_*\big(\operatorname{ch}(\alpha) \cdot \operatorname{Td}(p^*T_Y)^{-1} \cdot \operatorname{Td}(T_X)\big) \cdot \operatorname{Td}(T_Y) \\ &= p_*(\operatorname{ch}(\alpha) \cdot \operatorname{Td}(T_X)).\end{aligned}$$

□

Theorem 12.5 (Grothendieck–Riemann–Roch, general case)**.** *For any proper morphism $f : X \to Y$ of smooth varieties and $\alpha \in K_0(X)$:*

$$\mathrm{ch}(f_!\alpha) \cdot \mathrm{Td}(T_Y) = f_*(\mathrm{ch}(\alpha) \cdot \mathrm{Td}(T_X)).$$

Equivalently, the following diagram commutes:

$$\begin{tikzcd} K_0(X) \arrow[r, "f_!"] \arrow[d, "\mathrm{ch}(-)\ \mathrm{Td}(T_X)"'] & K_0(Y) \arrow[d, "\mathrm{ch}(-)\ \mathrm{Td}(T_Y)"] \\ \mathrm{CH}^*(X) \otimes \mathbb{Q} \arrow[r, "f_*"'] & \mathrm{CH}^*(Y) \otimes \mathbb{Q} \end{tikzcd}$$

Proof. Every proper morphism $f : X \to Y$ of smooth varieties factors as $f = p \circ i$ where $i : X \hookrightarrow X \times Y$ is the graph (a regular embedding by smoothness of f) and $p : X \times Y \to Y$ is the projection (smooth and proper). Apply Theorem 12.3 to i and Theorem 12.4 to p and compose. The compatibility of the Todd classes under composition follows from Lemma 12.2(ii) and the exact sequence $0 \to i^*T_{X\times Y/Y} \to T_X \to N_i \to 0$, giving $\mathrm{Td}(T_X) = \mathrm{Td}(i^*T_{X\times Y/Y}) \cdot \mathrm{Td}(N_i)^{-1}$. □

13. Chow's theorem: algebraicity of analytic sets

13.1. **Setup.** We work over $k = \mathbb{C}$. For a projective variety $X \subseteq \mathbb{P}^n_{\mathbb{C}}$, let X^{an} denote the underlying complex analytic space. A *closed analytic subset* $Z \subseteq \mathbb{P}^n(\mathbb{C})$ is a subset locally defined by the vanishing of finitely many holomorphic functions.

Theorem 13.1 (Chow 1949)**.** *Every closed analytic subset $Z \subseteq \mathbb{P}^n(\mathbb{C})$ is algebraic: there exists a homogeneous ideal $I \subseteq \mathbb{C}[x_0, \dots, x_n]$ with $Z = V(I)$.*

Proof. We give the complete proof using Cartan's theorem B and Hartogs' extension.

Step 1: Oka coherence. The ideal sheaf $\mathcal{I}_Z^{\mathrm{an}} \subseteq \mathcal{O}_{\mathbb{P}^n}^{\mathrm{an}}$ (the sheaf of holomorphic functions vanishing on Z) is a coherent analytic sheaf. This is *Oka's coherence theorem*: the sheaf of germs of holomorphic functions vanishing on a closed analytic set is coherent (proved by showing that the kernel of any holomorphic map between coherent sheaves is coherent, using the Weierstrass preparation theorem to establish coherence locally).

Step 2: Finiteness via Cartan–Serre. For each $d \geq 0$, consider the twisted sheaf $\mathcal{I}_Z^{\mathrm{an}}(d) = \mathcal{I}_Z^{\mathrm{an}} \otimes_{\mathcal{O}^{\mathrm{an}}} \mathcal{O}^{\mathrm{an}}(d)$. By the theorem of Cartan and Serre (1953): on a compact complex manifold (here $\mathbb{P}^n(\mathbb{C})$), every coherent analytic sheaf has finite-dimensional cohomology. In particular:

$$V_d := H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{I}_Z^{\mathrm{an}}(d)) \quad \text{is finite-dimensional over } \mathbb{C}.$$

For d sufficiently large, $\mathcal{I}_Z^{\mathrm{an}}(d)$ is generated by its global sections (this is the analytic analogue of Serre's vanishing theorem, proved by Cartan using the $\bar{\partial}$-equation and L^2 methods on Stein open sets).

Step 3: Polynomial nature of sections. A section $s \in V_d = H^0(\mathbb{P}^n(\mathbb{C}), \mathcal{I}_Z^{\mathrm{an}}(d))$ is a holomorphic section of $\mathcal{O}^{\mathrm{an}}(d)$ vanishing on Z. Via the quotient map $\pi : \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{P}^n(\mathbb{C})$, such a section pulls back to a holomorphic function $\tilde{s} : \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{C}$ that is:

- homogeneous of degree d: $\tilde{s}(\lambda z) = \lambda^d \tilde{s}(z)$ for $\lambda \in \mathbb{C}^\times$.
- vanishing on $\pi^{-1}(Z)$.

By Hartogs' extension theorem (a holomorphic function on $\mathbb{C}^{n+1} \setminus \{0\}$ extends to all of $\mathbb{C}^{n+1}$ when $n \geq 1$, since the singularity set has complex codimension ≥ 2), $\tilde{s}$ extends to a holomorphic function on all of $\mathbb{C}^{n+1}$.

A holomorphic function on $\mathbb{C}^{n+1}$ that is homogeneous of degree d is a polynomial. Indeed, expand in Taylor series: $\tilde{s}(z) = \sum_{|\alpha|\geq 0} a_\alpha z^\alpha$. Homogeneity forces $a_\alpha = 0$ for $|\alpha| \neq d$. So $\tilde{s} \in \mathbb{C}[z_0, \dots, z_n]_d$ is a homogeneous polynomial of degree d.

Step 4: Conclusion. Let $I = \bigoplus_{d\geq 0} V_d \subseteq \mathbb{C}[x_0, \ldots, x_n]$. This is a homogeneous ideal (closed under multiplication by polynomials, by the sheaf structure). We claim $Z = V(I)$.

$Z \subseteq V(I)$: by construction, every $s \in V_d$ vanishes on Z, so $Z \subseteq V(I)$.

$V(I) \subseteq Z$: for $d \gg 0$, $\mathcal{I}_Z^{\mathrm{an}}(d)$ is generated by its sections V_d (by Step 2). This means every holomorphic function vanishing on Z (locally on $\mathbb{P}^n$, twisted by d) is a linear combination of elements of V_d. Hence the zero locus of V_d (viewed as holomorphic functions on $\mathbb{P}^n$) is exactly Z. Thus $V(I) = Z$. □

Corollary 13.2. *Let $X \subseteq \mathbb{P}^n(\mathbb{C})$ be a projective algebraic variety. Every closed analytic subset $Z \subseteq X^{\mathrm{an}}$ is an algebraic subvariety of X.*

Proof. Since $X^{\mathrm{an}} \subseteq \mathbb{P}^n(\mathbb{C})$ is itself a closed analytic subset, and $Z \subseteq X^{\mathrm{an}}$ is closed in X^{an}, Z is also a closed analytic subset of $\mathbb{P}^n(\mathbb{C})$. By Theorem 13.1, $Z = V(J) \cap \mathbb{P}^n(\mathbb{C})$ for some homogeneous ideal J. Then $Z = V(J) \cap X = V(J + I(X))$ is an algebraic subvariety of X. □

14. Serre's GAGA theorem

14.1. **Statement and outline.** Let X be a projective scheme over $\mathbb{C}$ and let X^{an} be the associated complex analytic space. There is a natural morphism of ringed spaces $\epsilon : (X^{\mathrm{an}}, \mathcal{O}_{X^{\mathrm{an}}}) \to (X, \mathcal{O}_X)$, and for any coherent algebraic sheaf $\mathcal{F}$ on X, the analytification $\mathcal{F}^{\mathrm{an}} = \epsilon^*\mathcal{F}$ is a coherent analytic sheaf on X^{an}.

Theorem 14.1 (Serre, GAGA 1956)**.** *Let X be a projective variety over $\mathbb{C}$. Then:*

(i) For any coherent algebraic sheaf $\mathcal{F}$ on X and any $i \geq 0$, the natural comparison map
$$H^i(X, \mathcal{F}) \xrightarrow{\sim} H^i(X^{\mathrm{an}}, \mathcal{F}^{\mathrm{an}})$$
is an isomorphism of $\mathbb{C}$-vector spaces.

(ii) The analytification functor $\mathcal{F} \mapsto \mathcal{F}^{\mathrm{an}}$ is an equivalence of categories:
$$\{\text{coherent algebraic sheaves on } X\} \xrightarrow{\sim} \{\text{coherent analytic sheaves on } X^{\mathrm{an}}\}.$$

(iii) Every holomorphic map $f : X^{\mathrm{an}} \to Y^{\mathrm{an}}$ between analytifications of projective varieties is algebraic.

Proof (sketch of main steps). We outline the proof for $X = \mathbb{P}^n_{\mathbb{C}}$; the general case reduces to this.

Step 1: Cohomology comparison for $\mathcal{O}(d)$. For the algebraic line bundle $\mathcal{O}_{\mathbb{P}^n}(d)$:

- Algebraically: $H^0(\mathbb{P}^n, \mathcal{O}(d)) = \mathbb{C}[x_0, \ldots, x_n]_d$ (homogeneous polynomials of degree d), $H^i(\mathbb{P}^n, \mathcal{O}(d)) = 0$ for $0 < i < n$, and $H^n(\mathbb{P}^n, \mathcal{O}(d)) = \mathbb{C}[x_0, \ldots, x_n]^\vee_{-d-n-1}$ (Serre duality).
- Analytically: by the Dolbeault resolution, $H^i_{\mathrm{an}}(\mathbb{P}^n, \mathcal{O}^{\mathrm{an}}(d)) = H^{0,i}_{\bar\partial}(\mathbb{P}^n, \mathcal{O}(d))$, which can be computed explicitly and gives the same answer.

So (i) holds for $\mathcal{F} = \mathcal{O}(d)$.

Step 2: Induction on coherent sheaves. For a general coherent sheaf $\mathcal{F}$ on $\mathbb{P}^n$, use a *Hilbert syzygy resolution*: there exists an exact sequence
$$0 \to \bigoplus \mathcal{O}(-d_m) \to \cdots \to \bigoplus \mathcal{O}(-d_0) \to \mathcal{F} \to 0.$$
Analytifying gives an exact sequence of coherent analytic sheaves (analytification is exact since ϵ^* is right-exact and coherence ensures left-exactness of the cohomology comparison). The five-lemma applied to the long exact sequences in cohomology (algebraic and analytic) reduces the isomorphism for $\mathcal{F}$ to the isomorphism for $\mathcal{O}(d)$, proved in Step 1.

Step 3: Fully faithful (essential for (ii)). We must show that for coherent algebraic sheaves $\mathcal{F}, \mathcal{G}$ on X, the natural map
$$\mathrm{Hom}_X(\mathcal{F}, \mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{X^{\mathrm{an}}}(\mathcal{F}^{\mathrm{an}}, \mathcal{G}^{\mathrm{an}})$$

is an isomorphism. This follows from (i) with $\mathcal{H}om(\mathcal{F},\mathcal{G})$: $\mathrm{Hom}(\mathcal{F},\mathcal{G}) = H^0(X,\mathcal{H}om(\mathcal{F},\mathcal{G}))$ and $H^0(X^{\mathrm{an}},\mathcal{H}om^{\mathrm{an}}(\mathcal{F}^{\mathrm{an}},\mathcal{G}^{\mathrm{an}}))$, and both equal by Step 2.

Step 4: Essential surjectivity (main content of (ii)). Given a coherent analytic sheaf $\mathcal{G}$ on X^{an}, find a coherent algebraic $\mathcal{F}$ with $\mathcal{F}^{\mathrm{an}} \cong \mathcal{G}$. For $X = \mathbb{P}^n$: $\mathcal{G}(d) = \mathcal{G} \otimes \mathcal{O}^{\mathrm{an}}(d)$ is generated by global sections for $d \gg 0$ (Cartan's theorem A for $\mathbb{P}^n$ viewed analytically). So $\mathcal{G}$ is a quotient of $\bigoplus \mathcal{O}^{\mathrm{an}}(-d)$. The kernel is again coherent, and by a Hilbert basis argument (the Noetherian property of the analytic coherent sheaves on $\mathbb{P}^n$, proved by Cartan), we eventually obtain a presentation $\bigoplus \mathcal{O}^{\mathrm{an}}(-d') \to \bigoplus \mathcal{O}^{\mathrm{an}}(-d) \to \mathcal{G} \to 0$. Comparing with the algebraic map $\bigoplus \mathcal{O}(-d') \to \bigoplus \mathcal{O}(-d)$ (which exists by (i) applied to Hom groups) gives $\mathcal{G} \cong \mathrm{coker}^{\mathrm{an}}(\phi) = (\mathrm{coker}\phi)^{\mathrm{an}}$ for some algebraic map ϕ.

Step 5: Statement (iii). A holomorphic map $f : X^{\mathrm{an}} \to Y^{\mathrm{an}}$ corresponds to a sheaf map $f^{-1}\mathcal{O}_Y^{\mathrm{an}} \to \mathcal{O}_X^{\mathrm{an}}$, and the graph $\Gamma_f \subseteq X^{\mathrm{an}} \times Y^{\mathrm{an}} = (X \times Y)^{\mathrm{an}}$ is a closed analytic subset. By Chow's theorem (Theorem 13.1) applied to $X \times Y$ (which is projective), Γ_f is algebraic. The algebraic subvariety $\Gamma_f \subseteq X \times Y$ is the graph of an algebraic morphism $X \to Y$. □

15. Bloch's formula via the Gersten resolution

Theorem 15.1 (Bloch's formula)**.** *For a smooth variety X over a field k:*

$$\mathrm{CH}^p(X) \cong H^p_{\mathrm{Zar}}(X, \mathcal{K}_p)$$

where $\mathcal{K}_p$ is the Zariski sheaf $U \mapsto K_p(\mathcal{O}_X(U))$ (the p-th algebraic K-group functor).

Proof. The proof uses Quillen's computation of K-theory and his proof of the Gersten conjecture.

Step 1: Gersten resolution. For a smooth variety X of dimension n over a field, Quillen proved (1973) that for each $p \geq 0$ there is an exact sequence of sheaves (the *Gersten complex*):

$$0 \to \mathcal{K}_p \to \bigoplus_{x\in X^{(0)}} (i_x)_* K_p(k(x)) \xrightarrow{\partial} \bigoplus_{x\in X^{(1)}} (i_x)_* K_{p-1}(k(x)) \xrightarrow{\partial} \cdots \to \bigoplus_{x\in X^{(n)}} (i_x)_* K_{p-n}(k(x)) \to 0$$

where $X^{(j)}$ denotes the set of points of codimension j and $i_x : \{x\} \hookrightarrow X$ is the inclusion. The differentials ∂ are the tame symbol maps (boundary maps in the localization sequence for K-theory).

The exactness of this complex at the term $\bigoplus_{x\in X^{(j)}} (i_x)_* K_{p-j}(k(x))$ is the Gersten conjecture (proved by Quillen for smooth varieties over a field using his Q-construction and a homotopy equivalence argument for the K-theory of regular local rings).

Step 2: The complex is flasque. Each sheaf $\bigoplus_{x\in X^{(j)}} (i_x)_* K_{p-j}(k(x))$ is a direct sum of skyscraper sheaves, hence flasque (restriction to any open subset is surjective). Therefore the Gersten complex is a flasque resolution of $\mathcal{K}_p$.

Step 3: Cohomology computation. Since the Gersten complex is a flasque resolution:

$$H^q_{\mathrm{Zar}}(X, \mathcal{K}_p) = H^q\left(\Gamma\left(\bigoplus_{x\in X^{(\bullet)}} (i_x)_* K_{p-\bullet}(k(x))\right)\right).$$

For $q = p$: the complex $\Gamma(\bigoplus_x (i_x)_* K_{p-j}(k(x))) = \bigoplus_{x\in X^{(j)}} K_{p-j}(k(x))$ (global sections of a direct sum of skyscrapers). The p-th cohomology is:

$$H^p = \ker\left(\bigoplus_{x\in X^{(p)}} K_0(k(x)) \xrightarrow{\partial} \bigoplus_{x\in X^{(p+1)}} K_{-1}(k(x))\right) / \mathrm{im}\left(\bigoplus_{x\in X^{(p-1)}} K_1(k(x)) \xrightarrow{\partial} \bigoplus_{x\in X^{(p)}} K_0(k(x))\right).$$

Step 4: Identification with $\mathrm{CH}^p(X)$. We have:

- $K_0(k(x)) = \mathbb{Z}$ for each $x \in X^{(p)}$ (as $k(x)$ is a field), generated by the unit $[k(x)]$.
- $K_{-1}(k(x)) = 0$ for all x (negative K-groups of fields vanish).

- The image of $\partial : \bigoplus_{x\in X^{(p-1)}} K_1(k(x)) \to \bigoplus_{x\in X^{(p)}} K_0(k(x))$ is exactly the group of rational equivalences on codimension-p cycles.

Indeed, $K_1(k(x)) = k(x)^\times$ for each $x \in X^{(p-1)}$ (as the first K-group of a field is its multiplicative group, by Bass's calculation). The tame symbol map $\partial : k(x)^\times \to \bigoplus_{y\in\overline{\{x\}}^{(1)}} \mathbb{Z}$ sends $f \in k(x)^\times$ to $\sum_y \mathrm{ord}_y(f)\cdot[y]$, which is exactly the principal divisor $\mathrm{div}(f)$ on the closure $\overline{\{x\}} \subseteq X$.

Therefore:

$$H^p_{\mathrm{Zar}}(X,\mathcal{K}_p) = \frac{\bigoplus_{x\in X^{(p)}} \mathbb{Z}\cdot[x]}{\left\{\sum_y \mathrm{ord}_y(f)[y] : f\in k(x)^\times, x\in X^{(p-1)}\right\}} = \mathrm{CH}^p(X). \qquad \square$$

Corollary 15.2. *For $p = 1$: $\mathrm{CH}^1(X) \cong H^1(X,\mathcal{O}_X^\times) = \mathrm{Pic}(X)$. The Gersten resolution at $p = 1$ is the exact sequence*

$$0 \longrightarrow \mathcal{O}_X^\times \longrightarrow \mathcal{K}_1 = k(X)^\times \xrightarrow{\mathrm{div}} \bigoplus_{Y\ prime} \mathbb{Z}\cdot[Y] \longrightarrow 0$$

recovering the classical divisor theory.

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